Started on	Thursday, 4 December 2025, 1:19 PM
State	Finished
Completed on	Thursday, 4 December 2025, 1:29 PM
Time taken	9 mins 46 secs
Marks	25.00/25.00
Grade	**100.00** out of 100.00

Question 1

Complete

Mark 1.00 out of 1.00

$lookup performs poorly when:

- a. The foreign collection is sharded
- b. Both collections share the same shard key
- c. The local field is unique
- d. The join is performed on indexed fields

Question 2

Complete

Mark 1.00 out of 1.00

A MongoDB index with sparse: true will NOT index documents where:

- a. Index field exists but is empty
- b. Index field is null
- c. Index field does not exist
- d. Index field value is 0

Question 3

Complete

Mark 1.00 out of 1.00

A query using LIKE '%abc' cannot use a B-Tree index because:

- a. LIKE never uses indexes
- b. abc contains lowercase letters
- c. The wildcard appears at the end
- d. The wildcard appears at the beginning

Question 4

Complete

Mark 1.00 out of 1.00

If _id index exists, which situation will cause a COLLSCAN even when filtering on _id?

- a. _id contains strings
- b. Query uses $regex on _id
- c. Query uses _id: { $gte: 10 }
- d. Query uses _id: { $in: [1,2,3] }

Question 5

Complete

Mark 1.00 out of 1.00

In a sharded MongoDB cluster, the config server primarily stores:

- a. Chunk metadata and cluster configuration
- b. Write-ahead logs
- c. Index definitions
- d. All user data

Question 6

Complete

Mark 1.00 out of 1.00

In MongoDB, when using $facet, which of the following is TRUE?

- a. $facet can only be used on sharded clusters
- b. All pipelines inside $facet run sequentially
- c. $facet allows multiple pipelines to run on the same input in parallel
- d. $facet disables all indexes

Question 7

Complete

Mark 1.00 out of 1.00

In MongoDB, which of the following queries can use a compound index on { age: 1, score: -1 } most efficiently?

- a. db.users.find({ score: { $gt: 50 } }).sort({ age: 1 })
- b. db.users.find({ age: { $gte: 20 } }).sort({ age: 1 })
- c. db.users.find({ score: { $gte: 20 } }).sort({ score: 1 })
- d. db.users.find({ age: { $gte: 20 } }).sort({ score: -1 })

Question 8

Complete

Mark 1.00 out of 1.00

In MongoDB, which situation makes a compound index { a: 1, b: 1 } unusable for sorting?

- a. Query filters on a and sorts by { b: 1 }
- b. Query filters on both a and b
- c. Query sorts by { b: 1, a: 1 }
- d. Query sorts by { a: -1, b: 1 }

Question 9

Complete

Mark 1.00 out of 1.00

In MySQL InnoDB, a secondary index lookup requires:

- a. Reading only the secondary index
- b. Reading both secondary index and clustered index
- c. No lookup in clustered index
- d. A single B-tree traversal

Question 10

Complete

Mark 1.00 out of 1.00

MongoDB multi-document ACID transactions require which storage engine?

- a. WiredTiger
- b. InnoDB
- c. RocksDB
- d. MyISAM

Question 11

Complete

Mark 1.00 out of 1.00

MySQL performs a full table scan when:

- a. Partition key is present
- b. Query references a non-indexed column in WHERE
- c. Query returns fewer rows
- d. Index fits entirely in memory

Question 12

Complete

Mark 1.00 out of 1.00

To force MySQL to use a specific index, you use:

- a. FORCE THIS INDEX
- b. USE ONLY INDEX
- c. USE INDEX
- d. INDEX FORCE

Question 13

Complete

Mark 1.00 out of 1.00

Which aggregation operator allows you to reshape documents by controlling inclusion/exclusion of fields?

- a. $project
- b. $map
- c. $reduce
- d. $merge

Question 14

Complete

Mark 1.00 out of 1.00

Which isolation level in MySQL prevents dirty reads but still allows non-repeatable reads and phantom reads?

- a. READ UNCOMMITTED
- b. REPEATABLE READ
- c. READ COMMITTED
- d. SERIALIZABLE

Question 15

Complete

Mark 1.00 out of 1.00

Which MongoDB feature ensures that writes go to the primary node before being replicated?

- a. Journaling
- b. Write Concern w:1
- c. Write Concern w:majority
- d. Read Preference primaryPreferred

Question 16

Complete

Mark 1.00 out of 1.00

Which MongoDB write concern guarantees that data is written to majority of replica set nodes before acknowledging?

- a. w:all
- b. w:1
- c. w:majority
- d. journaled

Question 17

Complete

Mark 1.00 out of 1.00

Which MySQL condition forces the optimizer to avoid using an index even if one exists?

- a. column IN (1,2,3)
- b. WHERE column > 0
- c. WHERE function(column) = value
- d. column = 10

Question 18

Complete

Mark 1.00 out of 1.00

Which MySQL join returns rows that have matching values in both tables but excludes unmatched rows?

- a. FULL OUTER JOIN
- b. LEFT JOIN
- c. INNER JOIN
- d. RIGHT JOIN

Question 19

Complete

Mark 1.00 out of 1.00

Which MySQL storage engine does not support foreign keys?

- a. InnoDB
- b. NDB
- c. MyISAM
- d. MEMORY

Question 20

Complete

Mark 1.00 out of 1.00

Which of the following best describes EXPLAIN in MySQL?

- a. It executes the query with maximum optimization
- b. It updates statistics for the optimizer
- c. It rewrites queries automatically
- d. It shows the execution plan without running the query

Question 21

Complete

Mark 1.00 out of 1.00

Which of the following causes index intersection to be used?

- a. Query matches on two fields that each have separate single-field indexes
- b. Query uses $lookup
- c. Compound index exists
- d. Shard key is compound

Question 22

Complete

Mark 1.00 out of 1.00

Which of the following is true for clustered indexes in InnoDB?

- a. They store only pointers to rows
- b. They store full row data in the index
- c. They require manual configuration
- d. They can be disabled

Question 23

Complete

Mark 1.00 out of 1.00

Which operator allows you to execute pipeline stages for each document inside an array?

- a. $each
- b. $map
- c. $pipeline
- d. $unwind

Question 24

Complete

Mark 1.00 out of 1.00

Which query will lock the selected rows and prevent other transactions from reading them in InnoDB?

- a. SELECT ... FOR UPDATE
- b. SELECT ... AS LOCKED
- c. SELECT ... FROM ...
- d. SELECT ... FROM ... LOCK

Question 25

Complete

Mark 1.00 out of 1.00

Which SQL query guarantees eliminating duplicates before ordering the output?

- a. SELECT * FROM table ORDER BY col DISTINCT;
- b. SELECT ORDER DISTINCT * FROM table;
- c. SELECT DISTINCT * FROM table ORDER BY col;
- d. SELECT UNIQUE * FROM table ORDER BY col;